CHE110 · Lecture 2 study guide

Built from the full Lecture 2 transcripts (L2P1, L2P2, L2P3) and your Lecture 2–3 slides. TAUGHT IN CLASS said in the lecture · EXAM SIGNAL the lecturer said it is tested · SLIDES on the slides, not mentioned in the audio.

What the lecturer said about the exam

1. **"Whatever I taught today, this much is the syllabus for the midterm."** That means everything in this guide's class notes (sections 1–7) is midterm material, together with Lecture 1.
2. **"I am teaching everything that will come in the exam"**, including true/false, MCQ and short questions on the points below.
3. **Maximum / minimum wavelength** of the five series is a **common question** and "many people make mistakes". Do all five series and keep it in your notebook (section 2).
4. **Place the lower level first in the Rydberg formula.** Many students put the higher value in n_1. The lower value goes first.
5. **Memorise the series in order** (Lyman, Balmer, Paschen, Brackett, Pfund) so that the series name gives you n_1 immediately.
6. **Nodes**: radial / angular / total node calculations "appear in short questions and a lot in MCQs".
7. **Quantum numbers**: "found in short questions". For "which quantum number indicates orientation?" the symbol or name is enough ("never needs to be explained"). You must be able to say what each quantum number indicates.
8. **Possible orbital? Valid set?** ("Is 3f possible?", "Are these four quantum numbers correct?") appear as **true/false**. If a short reason is asked, give it **mathematically**.
9. **Electron configurations of atoms 1 to 30**. Not beyond 30. (Up to 118 exists but is not needed.) Know the exceptions.
10. **de Broglie true/false:** "Human beings do not have a wavelength" is **FALSE**.
11. Not required in depth: Schrödinger's equation ("do not memorise"), the probability graphs (not asked to draw), de Broglie's and Heisenberg's maths beyond what was explained. Understand the short statements "to the extent that I am explaining them".
12. **Exam tip:** there is enough time. Work step by step, use blank space for rough work, and don't cross out a correct answer (crossed-out answers are not marked).

Points the lecturer flagged

Max λ ↔ min energy ↔ nearest level | Min λ ↔ max energy ↔ from ∞

Lower level goes first (n_1) | de Broglie: λ = h/mv

Humans DO have a wavelength (false statement) | Heisenberg: position vs momentum

Node: probability zero | Radial node = n − l − 1 | Angular node = l | Total node = n − 1

1s has no node; s has no angular node | What each quantum number indicates

l: 0 to n−1 · m_l: −l to +l · m_s: ±½ | 3f, 2d, 1d not possible | Pauli · Aufbau (n+l) · Hund

Config of atoms 1–30 + exceptions

What was taught, in order

#	Topic	Core takeaway
1	Follow-ups from Lecture 1	Why ∞ → n_1 gives max energy gap; radius ↑ and velocity ↓ with n; c is the photon's speed; Bohr's circular orbit is wrong (elliptical, Sommerfeld).
2	Series revisited	All series are emission; max / min λ for each; lower limit fixed.
3	de Broglie	Dual nature; λ = h/mv; everything has a wavelength.
4	Heisenberg and Schrödinger	Uncertainty in position vs momentum; wave functions replace orbits; quantum numbers come from solving the wave equation.
5	Orbitals and nodes	Electron clouds; radial and angular nodes; formulas.
6	Quantum numbers	n, l, m_l, m_s: what each indicates and the allowed values.
7	Electron configuration	Aufbau, Pauli, Hund; exceptions; atoms 1–30.

1 · Bohr's model: follow-ups

TAUGHT IN CLASS

Energy at infinity, and the big energy gap

- At **n = ∞** the electron is **almost free**: it feels **no attraction** from the nucleus.

- For an electron falling from ∞ to a low level (1 or 2), the energy difference is the **maximum**, because you are subtracting from a level where it is essentially free.
- Close to the nucleus the attraction is strong; the first orbit's electron is the most strongly bound, so the gap to the free state is the biggest.
- **Easier to remove an electron from a higher level.** "We do not need to provide as much energy to remove an electron from a higher level, because the higher you go, the attraction decreases."

> **Two ways the word "energy" was used (read the question carefully).**
> • **Lecture 1:** the energy *of the level* increases with n ($E_1 < E_2 < \dots$, n = ∞ highest; in numbers $E = -13.6$ eV/n^2, rising from −13.6 toward 0).
> • **Lecture 2 (opening, answering a student):** the lecturer said the first-orbit electron has the "highest" energy and that energy "decreases" as you go higher. Here she was talking about the **binding energy** (how strongly the electron is held, the energy needed to free it), which is largest at n = 1 and falls with n. Both are consistent. **Level energy rises with n; binding / ionization energy falls with n.** If a question says "energy needed to remove / ionization / attraction", use the falling version. If it says "energy of the orbit", use the rising version. If a question is ambiguous, ask the lecturer which convention she wants.

Radius and velocity

- As you move farther from the nucleus, the **radius increases** and the **velocity decreases**. EXAM SIGNAL
- This comes from the **limitations of Bohr's model**.

Whose velocity is "c"?

- **c is the velocity of a photon**, not the electron. A photon is massless, so it travels at the speed of light.
- A photon comes in and excites an electron. When the electron drops back down it releases the same photon (same energy) it absorbed.
- **An electron's velocity is never c**; it is always less than the speed of light.

Circular or elliptical? (Sommerfeld)

- Bohr said the electron moves in a **circular** path, so the radius is constant and the velocity is fixed.
- In reality the electron moves in an **elliptical** path. In the same orbit it is sometimes very close to the nucleus (**velocity higher**) and sometimes far (**velocity lower**).
- **Sommerfeld** later explained this and corrected many aspects of Bohr's model (the Sommerfeld model).

Is angular momentum still quantized? (the 5 × 10 = 10 × 5 idea)

In an elliptical orbit r changes and v changes. If r gets smaller, v gets larger, and the product stays the same (like 5 × 10 = 10 × 5 = 50). Mass is constant. So **mvr stays fixed**, which proves the angular-momentum quantization is correct. **But Bohr only gave the formula.** His circular-path picture is wrong, and he gave no logical explanation. Scientists saw that the formula worked but wouldn't fully accept it without an explanation. **de Broglie** supplied that explanation (section 3).

2 · The series: maximum and minimum wavelength

TAUGHT IN CLASS COMMON EXAM QUESTION

Spectrum recap

- An electron absorbs a photon's energy → **absorption** (moves up). When it drops from a higher level to a lower one it emits the photon → **emission** (moves down).
- The radiation we usually observe belongs to **emission spectra**. All five series (Lyman, Balmer, Paschen, Brackett, Pfund) are **emission** (high → low), each with a **fixed lower limit**.
- If the series name is given, you should immediately know n_1.

Series order and n_1: Lyman 1 · Balmer 2 · Paschen 3 · Brackett 4 · Pfund 5.

The rules

Case	Transition	Energy	Wavelength
Maximum energy	From **n = ∞** down to n_1	Max	**Minimum λ**
Minimum energy	From the **immediate next level** (n_1 + 1) to n_1	Min	**Maximum λ**

Energy and wavelength are inversely proportional, so learn the transition by **either** energy **or** wavelength and derive the other.

Common mistake (lecturer): even after the series is given, students put the **higher** value in n_1. **The lower level goes first.** A series means the scientist fixed a lower limit; the electron can come from any higher level, from the immediate one all the way up to infinity.

Table for your notebook ($R_H = 1.097\times10^{7}$ m^{-1}, $h = 6.626\times10^{-34}$ J·s, $c = 3\times10^{8}$ m/s)

Series	n_1	Max λ ($n_1+1 \to n_1$)	Min λ ($\infty \to n_1$)	Min energy	Max energy
Lyman	1	1.215×10^{-7} m	9.116×10^{-8} m	1.635×10^{-18} J	2.181×10^{-18} J
Balmer	2	6.563×10^{-7} m	3.646×10^{-7} m	3.029×10^{-19} J	5.452×10^{-19} J
Paschen	3	1.875×10^{-6} m	8.204×10^{-7} m	1.060×10^{-19} J	2.423×10^{-19} J
Brackett	4	4.051×10^{-6} m	1.459×10^{-6} m	4.906×10^{-20} J	1.363×10^{-19} J
Pfund	5	7.458×10^{-6} m	2.279×10^{-6} m	2.665×10^{-20} J	8.722×10^{-20} J

Worked: Balmer min λ = $4/R_H$ = 3.65×10^{-7} m; Lyman max λ = $4/(3R_H)$ = 1.215×10^{-7} m (the first two problems done on the board; "number 25" on your sheet). Others on your sheet: 4 → 2 (λ = 4.86×10^{-7} m) and Paschen 5 → 3 (λ = 1.282×10^{-6} m).

Use the calculator in the Lecture 1 guide for any transition.

3 · de Broglie: dual nature

TAUGHT IN CLASS TRUE / FALSE SIGNAL

- de Broglie proposed that **electrons (and other particles) have a dual nature**: particle (they have mass) and wave. Light was already known to be dual.
- He logically supported Bohr's formula using this idea (the wave explanation of quantized orbits).

$$\lambda = h / (m\,v)$$

- **mv is ordinary momentum** (mass × velocity), **not** angular momentum.
- High velocity → high frequency → **very small wavelength**. Low velocity → large wavelength.
- **Everything has a wavelength.** A 50 kg person running at 100 m/s: λ ≈ 6.626×10^{-34} / (50 × 100) ≈ **10^{-37} m**. This is **negligible** (can't be measured or seen), but **not non-existent**. A grain of sand and a multi-ton elephant also have one.
- The wavelength of **electrons, protons and neutrons is measurable**, because their mass is tiny. Example: electron m ≈ 10^{-31} kg, v ≈ 2.8×10^{8} m/s → λ ≈ 10^{-12} m (10^{-34} in the numerator over 10^{-31} in the denominator gives a difference of 10^{3}).

> **True/false trap:** "According to de Broglie, human beings do not have a wavelength." → **FALSE.** We have one; it's negligible. Nobody said it doesn't exist.

Use λ = h/mv with h = 6.626×10^{-34} J·s.

The anecdote

de Broglie's PhD thesis was **one page**: electrons, like light, show both particle and wave nature. The board couldn't decide and asked Einstein, who replied that the student knew more than they did. The PhD was accepted. (Told to show the idea is simple but wasn't easy to arrive at.)

4 · Heisenberg and Schrödinger

TAUGHT IN CLASS

Does the electron show both natures at once?

No. It shows one extreme at a time, and the other is suppressed (like a person showing only their good side at a time). This uncertainty is what **Heisenberg** explained.

Heisenberg's uncertainty principle

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

- x = **position**, p = **momentum** (mass × velocity).
- Think of the "=" version: if one is very large, the other must be small.
- If the **position is fixed (certain)**, you **cannot determine the velocity**. If you measure the **velocity accurately**, you cannot fix the **position**. When one becomes certain, the other becomes uncertain.
- Mass is treated as constant (relativistic mass change is not applied here).

> **Analogy used:** a moving object. If you can pinpoint its exact location, its velocity must be low (it is nearly still); if it is moving fast you can't pinpoint where it is.

Schrödinger and quantum mechanics

- The modern theory is **quantum mechanics**, based on **Schrödinger's wave equation**.
- Instead of saying an electron jumps from "orbit ∞ to orbit 2", we say it moves from **one wave-function (ψ) state to another**. The word **orbit is no longer needed**; you use **wave functions and mathematical equations**.
- The equation contains mass, Planck's constant, total energy and potential energy. You only need to know that it exists; **do not memorise it**. Solving it takes calculus (differentiation, integration).
- **Solving it gives values:** the probability of finding the electron at a location, its position, direction and how it moves along an axis. **These values are the quantum numbers** (n, l, m_l, m_s).

- For a **single electron** (e.g. H) it can be solved. With **many electrons** it can no longer be solved easily.
- Slide: ψ^2 gives the probability of finding the particle in a region of space. SLIDES
- The combination of **de Broglie, Heisenberg and Schrödinger** explains an electron's nature and gives the modern atomic model.

5 · Orbitals, electron clouds and nodes

TAUGHT IN CLASS MCQ / SHORT-QUESTION SIGNAL

Orbit vs orbital (analogy)

- An **orbit** is like a **building** (or road); an **orbital** is like the specific **flat or room** in it.
- Electrons exist in orbitals, never in the spaces between them. Most of the atom is empty space (Rutherford), which is why atomic radii are large.
- Electrons are not on thin lines. An electron exists as a **cloud**; the next level has its own cloud.

What is a node?

> The thin regions between clouds are like edges of two waves meeting with **destructive interference**. Electrons are found where waves match, not where they cancel. A **node** is a region where the **probability of finding an electron is (almost) zero**. You will never find an electron there.

Two types of node

	Radial node	**Angular node**
Meaning	A spherical region at a particular **distance (radius) from the nucleus** where probability = 0 (e.g. the white gap inside a 2s picture)	A **plane** through the nucleus where probability = 0 (between the positive and negative lobes of a p orbital)
Formula	**n − l − 1**	**l**
Total	**n − 1** (radial + angular)	

- **s orbital:** spherical, equal in x, y, z. There is **no phase change, so no angular node** (l = 0). It can only have radial nodes.
- **p, d, f** have lobes with positive and negative phase, so they have **angular nodes (= l)**.
- **1s has no node at all**: it is the closest to the nucleus, and there's nothing between it and the nucleus. n − l − 1 = 1 − 0 − 1 = 0; l = 0. Nodes only appear at higher levels.

- **2s:** 1 radial node, 0 angular. **3s:** 2 radial. **3p** (the picture): 1 radial + 1 angular (the line through the middle) = 2 total.
- The nodes increase as you go further, and the cloud density spreads out.

Radial = n − l − 1; angular = l; total = n − 1.

Orbital	n	l	Radial (n−l−1)	Angular (l)	Total (n−1)
1s	1	0	0	0	0
2s	2	0	1	0	1
2p	2	1	0	1	1
3s	3	0	2	0	2
3p	3	1	1	1	2
3d	3	2	0	2	2
4d **(class example)**	4	2	**1**	**2**	**3**

> **Class example, 4d:** total = n − 1 = 3. Angular = l = 2. Radial = 4 − 2 − 1 = 1. Check: 2 + 1 = 3.

Radial probability curves (to understand, not to draw)

The graph has distance on the x-axis and probability on the y-axis. **1s:** never touches zero (no node). **2s:** touches zero once (1 node). **3s:** touches zero twice (2 nodes). Where the curve hits zero is a node at that radius. The lecturer said you are not asked to draw these.

> **Aside (1s and electron capture).** 1s electrons are so close to the nucleus that the nucleus can sometimes capture one (**electron capture**). An electron from a higher level then drops into the vacancy and releases energy (the same idea as X-rays). Mentioned in passing; not an exam point.

6 · The four quantum numbers

TAUGHT IN CLASS EXAM SIGNAL

Quantum numbers come from solving the wave equation. Together they give each electron a unique identity.

Symbol	Name	What it indicates	Allowed values	Analogy
n	Principal	**Energy level (shell)**; K, L, M, N… for 1, 2, 3, 4…	1, 2, 3, …	The building (house number)
l	Azimuthal / subsidiary	**Orbital name / shape** (s, p, d, f)	**0 to n − 1** (0 = s, 1 = p, 2 = d, 3 = f)	The flat and its shape
m_l	Magnetic	**Orientation (direction) in 3D space**	**−l to +l** (2l + 1 values), integers only	Which room, in which direction
m_s	Spin	**Spin of the electron on its own axis** (clockwise / counter-clockwise)	**+½ or −½**	How you are positioned inside the room

- Numbers can be written as symbols too: n as K, L, M, N; l as s, p, d, f (easier to see the shape).
- **l → shape:** s spherical, p dumbbell, d double dumbbell.
- **m_s**: in one orbital (one direction), at most **two electrons**: one spins one way, the other the opposite way. The ± sign is the direction relative to a reference. (½ is explained as one possibility out of two: clockwise or counter-clockwise. You don't need the deeper physics.)
- The ± sign of m_l is the direction relative to the z-axis. **z is the reference, so $m_l = 0$ is the z orbital.**

> **Short-question tip:** "Which quantum number indicates orientation in 3D space?" → **magnetic (m_l)**. Write exactly what is asked; no explanation needed.

Possible values for n = 3 (class example)

- l = 0 to n − 1 = 0, 1, 2 → s, p, d (**3 values of l**).
- m_l for s: 0 → **1** value. For p: −1, 0, +1 → **3**. For d: −2 … +2 → **5**. Total m_l values = 1 + 3 + 5 = **$9 = n^2$** (shortcut).
- Max electrons in the shell = $2n^2 = 18$.
- The lecturer: if unsure of a shortcut, do it step by step.

Which orbital gets which m_l?

Subshell	m_l values	Orbitals (class order)
s	0	s (equal in all directions)
p	−1, 0, +1	p_x, p_z, p_y: **0 is p_z** (z is the reference); the other two are p_x and p_y
d	−2, −1, 0, +1, +2	d_xy, d_yz, **d_{z^2}**, d_xz, $d_{x^2-y^2}$: **0 is d_{z^2}**

The transcript is unclear on which of p_x/p_y takes −1 versus +1 (the lecturer put x first). The part to remember: **z / z^2 is 0**, and the rest are listed in series (xy, yz, [z^2], xz, x^2-y^2 for d).

> **Class question:** "n = 4, l = 2, m_l = 0: which orbital?" → **4d_z^2**. (n = 4 → shell; l = 2 → d; m_l = 0 → z^2.) If four quantum numbers are given, you can write the orbital notation.

Is the orbital possible? (true/false)

Orbital	Possible?	Reason (write it mathematically)
3f	No	l = 3 needs n ≥ 4 (l goes 0 to n − 1 = 2 for n = 3)
1d, 1p	No	n = 1 gives l = 0 only → only s
2d	No	n = 2 gives l = 0, 1 → s, p only
2p, 3d	Yes	l ≤ n − 1

Are these four quantum numbers valid? (true/false)

n = 4, l = 2, m_l = +3, m_s = +½.
n = 4 → l can be 0, 1, 2, 3 → l = 2 is allowed. But l = 2 → m_l must be between −2 and +2 → **+3 is not allowed**. So the set is **invalid**, and the statement "this set is valid" is false. Check them one by one: n, then l, then m_l, then m_s.

SLIDES Same method for the slide questions: n = 0 is invalid (n ≥ 1); n = 3, l = 2, m_l = +3 is invalid.

7 · Electron configuration

TAUGHT IN CLASS EXAM SIGNAL: atoms 1 to 30

The electron configuration tells you where each of the electrons sits. The 118 elements are explained with the four orbital types s, p, d, f (g and h are not known to exist yet). Example: sodium, **$1s^2$ $2s^2$ $2p^6$ $3s^1$** (11 electrons): first fill s, then p, then 3s in the next shell. It is not random.

Aufbau principle ("build-up")

- Electrons fill from **low energy to high energy**. The energy order of subshells is **s < p < d < f**. s is lowest because s electrons stay close to the nucleus: higher attraction and **penetration**, so electrons enter s first.
- **(n + l) rule:** the orbital with the **lower n + l** fills first. If n + l is equal, the one with the **lower n** fills first.

- Example: 4s (4 + 0 = 4) fills before 3d (3 + 2 = 5), even though 3d is in a lower shell.

$$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \ldots$$

Pauli exclusion principle

- **No two electrons in an atom can have the same set of all four quantum numbers.** The quantum numbers are each electron's unique identity (like names for twins).
- Two electrons in the same orbital (e.g. 2p_x): n, l, m_l are the same, so **m_s must differ** (+½ and −½). Hence an orbital holds **maximum two electrons, with opposite spins**.
- If they were identical, they would repel and the atom wouldn't exist. Nature stays balanced.

Hund's rule

- **Degenerate orbitals** = orbitals of the same energy: p has 3, d has 5, f has 7.
- Electrons enter degenerate orbitals **one by one first**, all with the **same spin**. Pairing starts only after every orbital has one electron (siblings and chocolates analogy: give one to each so nobody fights).
- Reason: a half-filled set is a **symmetric** structure with less repulsion. **Half-filled and fully-filled** sets are **stable**.
- The first electron may go into **any** of the degenerate orbitals (p_x, p_y or p_z); there's no rule about which first.
- Examples (slides): N (7) = $1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$. O (8) = $1s^2\ 2s^2\ 2p_x^2\ 2p_y^1\ 2p_z^1$ (the fourth p electron pairs up in p_x with opposite spin).

Exceptions: half- and full-filled stability

> If $4s^2\ 3d^9$ were the configuration, moving one electron from 4s into 3d gives **$3d^{10}$ (full)**, which is more stable. So **copper (29) is $4s^1\ 3d^{10}$**, not $4s^2\ 3d^9$. Likewise **chromium (24) is $4s^1\ 3d^5$** (half-filled d) instead of $4s^2\ 3d^4$. The lecturer's rule: "half-filled and full-filled are stable". Atoms 1–30 contain these exceptions.

Z	Atom	Configuration
3	Li	$1s^2\ 2s^1$
6	C	$1s^2\ 2s^2\ 2p^2$
7	N	$1s^2\ 2s^2\ 2p^3$
8	O	$1s^2\ 2s^2\ 2p^4$
10	Ne	$1s^2\ 2s^2\ 2p^6$
11	Na	$1s^2\ 2s^2\ 2p^6\ 3s^1$
18	Ar	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$
19	K	$\ldots 3p^6\ 4s^1$
20	Ca	$\ldots 3p^6\ 4s^2$
21	Sc	$\ldots 4s^2\ 3d^1$
24	Cr (exception)	$\ldots 4s^1\ 3d^5$
29	Cu (exception)	$\ldots 4s^1\ 3d^{10}$
30	Zn	$\ldots 4s^2\ 3d^{10}$

Use the builder for all 30 and check yourself. (Some books write 3d before 4s in the final string; the filling order is the same.)

Slides extras (Lecture 2–3)

SLIDES Points on the slides that the lecturer didn't spend time on in the audio. Quiz-ready, but lower priority than the class notes above.

- **Orbit:** electrons move in two-dimensional, spherical, fixed energy levels called orbits; K (nearest the nucleus), L, M, N…
- **Orbital:** the 3D region where the probability of finding the electron is maximum (**90–95%**). s, p, d, f; **max two electrons per orbital**. The idea of orbitals came from the wave nature of electrons (not part of Bohr's theory). Atomic orbital model = electron cloud / wave mechanics model.
- **Orbit vs orbital:** orbit is a fixed circular path (Bohr, 2D, particle picture). Orbital is a 3D probability region (wave picture).
- **Shapes:** s spherical; p dumbbell (three: p_x, p_y, p_z, same energy, along perpendicular axes); d: five orbitals; f: seven orbitals.

- **Balmer** proposed the formula for the wave number of spectral lines in 1885; the **Rydberg formula** relates the energy difference between Bohr levels to the wavelengths of absorbed/emitted photons.
- **Spectrum:** Newton showed sunlight is made of colours. An incandescent bulb gives a continuous spectrum (all wavelengths). Emission: heated atoms emit certain wavelengths (bright lines). Absorption: white light through a substance leaves dark lines.
- **Bohr slide statements:** the first model with quantum energy levels; won the Nobel Prize in 1922; closer orbits (lower energy) are more stable; an electron in the lowest energy orbit is in the **ground state**; absorbing energy to a higher level gives an **excited state**.
- **Bohr limitations (slides):** explains only single-electron species; cannot explain fine structure (a line splits into several close lines → sub-levels → new quantum numbers); cannot explain splitting in magnetic or electric fields; the idea of electrons jumping between orbits conflicts with the uncertainty in position and velocity; circular vs elliptical.
- **Link:** 3D images of nodes at chemtube3d.com (3p orbitals).

Worked problems

1. Radial, angular and total nodes in 4d. n = 4, l = 2. Radial = 4 − 2 − 1 = 1; angular = 2; total = 3.

2. Nodes in 5f. n = 5, l = 3. Radial = 5 − 3 − 1 = 1; angular = 3; total = 4.

3. Nodes in 1s. Radial = 1 − 0 − 1 = 0; angular = 0. No node.

4. n = 3: possible l, m_l. l = 0, 1, 2; m_l count = 1 + 3 + 5 = 9 (= n^2).

5. Orbital from n = 4, l = 2, m_l = 0. 4d_z^2.

6. Is (n = 3, l = 3, m_l = 0, m_s = +½) valid? No: l must be 0 to n − 1 = 2.

7. Is (n = 2, l = 1, m_l = −1, m_s = −½) valid? Yes: l ≤ 1; m_l between −1 and +1; m_s = ±½. It is a 2p electron.

8. de Broglie, 50 kg at 100 m/s. $\lambda = 6.626\times10^{-34} / (50 \times 100) = 1.3\times10^{-37}$ m. Negligible, not zero.

9. de Broglie, electron. $m = 9.109\times10^{-31}$ kg, $v = 2.8\times10^{8}$ m/s: $\lambda = 6.626\times10^{-34} / (9.109\times10^{-31} \times 2.8\times10^{8}) = 2.6\times10^{-12}$ m (measurable).

10. Configuration of sodium (11). $1s^2\ 2s^2\ 2p^6\ 3s^1$. Last electron: $n = 3$, $l = 0$, $m_l = 0$, $m_s = +½$.

11. Why does 4s fill before 3d? $n + l$: $4s = 4 + 0 = 4$; $3d = 3 + 2 = 5$. Lower $n + l$ first.

12. Oxygen (8) by Hund. $1s^2\ 2s^2\ 2p^4$: the first three p electrons go singly into p_x, p_y, p_z (same spin); the fourth pairs with opposite spin in one of them.

13. Series check. Balmer min λ: $\infty \rightarrow 2$: $4/R = 3.65\times10^{-7}$ m. Lyman max λ: $2 \rightarrow 1$: $4/(3R) = 1.215\times10^{-7}$ m.

Flashcards

Question	Answer
Why is $\infty \rightarrow n_1$ the maximum-energy transition?	At ∞ the electron is almost free (no attraction); the gap to a tightly bound low level is the largest.
Maximum λ in a series comes from?	The immediate next level ($n_1 + 1 \rightarrow n_1$): minimum energy gap.
Minimum λ in a series comes from?	$n = \infty \rightarrow n_1$: maximum energy gap.
Which value goes first (n_1) in the Rydberg formula?	The lower level. A common mistake is to put the higher one first.
Series and their n_1?	Lyman 1, Balmer 2, Paschen 3, Brackett 4, Pfund 5.
Are the five series emission or absorption?	Emission (transition high → low), each with a fixed lower limit.
As n increases, what happens to radius and velocity?	Radius increases, velocity decreases.
Whose velocity is c?	The photon's. The electron's speed is always below c.
Does a photon have mass?	No, it is massless and travels at the speed of light.
What did Bohr get wrong about the path?	He said circular; it's elliptical (velocity varies within the orbit). Sommerfeld corrected it.
Is angular momentum still fixed on an elliptical path?	Yes: when r decreases v increases, so mvr is constant ($5\times10 = 10\times5$).
Who explained why Bohr's formula was right?	de Broglie, using dual (wave–particle) nature.
de Broglie equation?	$\lambda = h/(mv)$, where mv is ordinary momentum.
True or false: humans have no wavelength (de Broglie).	FALSE. It's about 10^{-37} m: negligible but existent.
Why can we measure electron wavelengths but not ours?	Electron mass is tiny (10^{-31} kg), so λ is measurable.
Heisenberg's uncertainty principle?	$\Delta x \cdot \Delta p \geq h/4\pi$: fixing position makes velocity (momentum) uncertain and vice versa.
What does Schrödinger's equation give?	Wave functions ψ; solving them gives probabilities and the quantum numbers (for 1 electron).

Does quantum mechanics use 'orbits'?	No. It uses wave functions and orbitals.
What is a node?	A place where the probability of finding the electron is zero.
Radial node formula?	$n - l - 1$
Angular node formula?	l
Total nodes?	$n - 1$
Nodes in 1s?	None.
Does an s orbital have angular nodes?	No ($l = 0$, spherical, no phase change).
Nodes in 4d?	Radial 1, angular 2, total 3.
What does n indicate?	Energy level (shell).
What does l indicate?	Orbital shape/name (s, p, d, f).
What does m_l indicate?	Orientation in 3D space.
What does m_s indicate?	Spin of the electron on its own axis (+½ / −½).
Allowed l?	0 to $n - 1$.
Allowed m_l?	$-l$ to $+l$ ($2l + 1$ values).
How many m_l values for $n = 3$?	9 ($= n^2$).
Is 3f possible? 2d? 1p?	No, no, no: l can't exceed $n - 1$.
Which m_l is the z orbital?	0 (z is the reference).
$n = 4$, $l = 2$, $m_l = 0$: which orbital?	$4d_{z^2}$.
Pauli exclusion?	No two electrons in an atom share all four quantum numbers (max 2 per orbital, opposite spins).
Aufbau principle?	Fill the lowest energy first; lowest $(n + l)$, ties go to lower n.
Why 4s before 3d?	$n + l$: $4s = 4 < 3d = 5$.
Hund's rule?	Fill degenerate orbitals singly (same spin) before pairing.
What are degenerate orbitals?	Orbitals with the same energy (3 p, 5 d, 7 f).
Which configurations are extra stable?	Half-filled and fully-filled subshells.
Configuration of Cr (24) and Cu (29)?	Cr: $4s^1\ 3d^5$. Cu: $4s^1\ 3d^{10}$ (exceptions).
Configuration of Na?	$1s^2\ 2s^2\ 2p^6\ 3s^1$.

Quiz & mixed review

Questions 1–44 are Lecture 2; the last questions mix in Lecture 1 and the slides.

1. For which transition is the energy difference maximum?
   A) $n_1 + 1 \rightarrow n_1$ B) $\infty \rightarrow n_1$ C) $n_1 \rightarrow \infty$ D) Always $2 \rightarrow 1$
   Answer: B. The electron at ∞ is free; the gap to the lower limit is the largest.
2. The minimum wavelength of a series comes from
   A) The immediate next level B) $n = \infty$ C) n_1 itself D) Random levels
   Answer: B. Max energy means min λ.
3. The maximum wavelength of a series comes from
   A) $n = \infty$ B) The immediate next level C) The lowest level D) Level 10
   Answer: B. Min energy means max λ.
4. In the Rydberg formula n_1 is
   A) The higher level B) The lower level C) Always 1 D) The series' last line
   Answer: B. Lower level first. A common mistake is to put the higher one first.
5. The Paschen series has n_1 =
   A) 2 B) 3 C) 4 D) 5
   Answer: B. Lyman 1, Balmer 2, Paschen 3, Brackett 4, Pfund 5.
6. All five hydrogen series are
   A) Absorption spectra B) Emission spectra C) Continuous spectra D) Band spectra
   Answer: B. High → low transitions.
7. Moving away from the nucleus, the electron's
   A) Radius ↓, velocity ↑ B) Radius ↑, velocity ↓ C) Both ↑ D) Both unchanged
   Answer: B. Radius increases, velocity decreases.
8. In $\Delta E = hc/\lambda$, c is the velocity of
   A) The electron B) The photon C) The nucleus D) Sound
   Answer: B. A massless photon travels at the speed of light.
9. The electron's velocity is
   A) Equal to c B) Always less than c C) Greater than c D) Zero
   Answer: B. Only the massless photon moves at c.
10. Bohr's circular path is incorrect because the real path is
    A) Straight B) Elliptical C) Square D) Random
    Answer: B. This was corrected by Sommerfeld.
11. Who later corrected Bohr's model for elliptical orbits?
    A) Sommerfeld B) Planck C) Schrödinger D) Pauli
    Answer: A. The Sommerfeld model.

12. Who explained why Bohr's formula was right using dual nature?
    A) Heisenberg B) de Broglie C) Hund D) Aufbau
    Answer: B. de Broglie: particles have wave nature.
13. The de Broglie relation is
    A) $\lambda = hc/E$ B) $\lambda = h/mv$ C) $\lambda = mv/h$ D) $\lambda = h \cdot mv$
    Answer: B. mv is ordinary momentum.
14. True or false: according to de Broglie, humans have no wavelength.
    A) True B) False
    Answer: B. Humans have one ($\sim 10^{-37}$ m) but it's negligible.
15. Why are electron wavelengths measurable?
    A) High charge B) Very small mass C) Speed of light D) They are photons
    Answer: B. Small m makes $\lambda = h/mv$ large enough.
16. Heisenberg's principle says
    A) Energy is quantized B) Position and momentum can't both be known exactly C) Electrons have spin D) Orbits are circular
    Answer: B. $\Delta x \cdot \Delta p \geq h/4\pi$.
17. If the position of an electron is known exactly, its velocity is
    A) Exactly known too B) Uncertain C) Zero D) Equal to c
    Answer: B. One becomes certain, the other uncertain.
18. Quantum mechanics is based on
    A) Dalton's laws B) Schrödinger's wave equation C) Rutherford's model D) Maxwell's equations
    Answer: B. It uses wave functions instead of orbits.
19. The quantum numbers come from
    A) Solving the wave equation B) Measuring orbits with a ruler C) Dalton's postulates D) The periodic table
    Answer: A. Wave-function solutions.
20. A node is
    A) Where the electron density is maximum B) Where the probability of finding the electron is zero C) The nucleus D) A photon
    Answer: B. Destructive interference regions.
21. The radial-node formula is
    A) l B) $n - 1$ C) $n - l - 1$ D) $n + l$
    Answer: C. Angular = l; total = $n - 1$.
22. How many total nodes does a 4d orbital have?
    A) 1 B) 2 C) 3 D) 4
    Answer: C. $n - 1 = 3$ (radial 1, angular 2).
23. How many radial nodes in 4d?
    A) 0 B) 1 C) 2 D) 3

Answer: B. 4 − 2 − 1 = 1.

24. Does 1s have a node?

A) Yes, one radial B) Yes, one angular C) No node D) Two nodes

Answer: C. There's nothing between 1s and the nucleus.

25. Which orbital type has no angular node?

A) s B) p C) d D) f

Answer: A. $l = 0$ and there's no phase change.

26. Which quantum number indicates orientation in 3D space?

A) n B) l C) m_l D) m_s

Answer: C. The magnetic quantum number.

27. l indicates

A) Energy level B) Orbital shape / name C) Orientation D) Spin

Answer: B. s, p, d, f.

28. n indicates

A) Energy level B) Spin C) Orientation D) Shape

Answer: A. Principal quantum number.

29. m_s can be

A) 0 or 1 B) +½ or −½ C) −1, 0, +1 D) Any integer

Answer: B. Spin up or down.

30. For n = 3, the values of l are

A) 1, 2, 3 B) 0, 1, 2 C) 0, 1, 2, 3 D) −2 to +2

Answer: B. 0 to n − 1.

31. For l = 2, m_l takes

A) 0, 1, 2 B) −2, −1, 0, +1, +2 C) −1, 0, +1 D) Only 0

Answer: B. −l to +l: five values (the d orbitals).

32. The total number of m_l values for n = 3 is

A) 3 B) 5 C) 9 D) 18

Answer: C. $1 + 3 + 5 = 9 = n^2$.

33. Which orbital is NOT possible?

A) 2p B) 3d C) 2d D) 4f

Answer: C. n = 2 only allows l = 0, 1 (s, p).

34. Why is 3f impossible?

A) f needs l = 3 and n = 3 allows l up to 2 B) 3f is too heavy C) n = 3 has no p D) f doesn't exist

Answer: A. $l \leq n - 1$.

35. n = 4, l = 2, $m_l = 0$ describes

A) 4p B) $4d_{z^2}$ C) $3d_{z^2}$ D) 4s

Answer: B. n = 4, d, $m_l = 0 \rightarrow z^2$.

36. Is ($n = 4, l = 2, m_l = +3, m_s = +½$) valid?

A) Yes B) No: m_l must be −2 … +2 C) No: n is too large D) No: m_s is wrong

Answer: B. For $l = 2$, $m_l \leq 2$.

37. Pauli exclusion principle says

A) Fill singly before pairing B) No two electrons share all four quantum numbers C) Lowest energy first D) Orbits are circular

Answer: B. Two electrons in an orbital must differ in spin.

38. Which rule fills orbitals in order of increasing (n + l)?

A) Pauli B) Aufbau C) Hund D) Heisenberg

Answer: B. Aufbau ('build-up').

39. Which fills first: 4s or 3d?

A) 3d B) 4s C) Same time D) Depends on the atom

Answer: B. 4s: $n + l = 4$; 3d: $n + l = 5$.

40. Hund's rule concerns

A) Degenerate orbitals B) Nodes C) Photons D) Radius

Answer: A. Fill singly with the same spin first.

41. How many degenerate orbitals are in a d subshell?

A) 1 B) 3 C) 5 D) 7

Answer: C. p has 3, d has 5, f has 7.

42. Which is a stable arrangement?

A) One-third-filled B) Half-filled or fully-filled C) Any partly filled D) Empty

Answer: B. Symmetric structure, less repulsion.

43. The configuration of Cu (29) is

A) …$4s^2\ 3d^9$ B) …$4s^1\ 3d^{10}$ C) …$4s^0\ 3d^{11}$ D) …$4s^2\ 3d^{10}$

Answer: B. A full d subshell is more stable.

44. The configuration of Na (11) is

A) $1s^2\ 2s^2\ 2p^6\ 3s^1$ B) $1s^2\ 2s^2\ 2p^5\ 3s^2$ C) $1s^2\ 2s^1\ 2p^6\ 3s^2$ D) $1s^2\ 2s^2\ 2p^7$

Answer: A. 11 electrons.

45. [L1] Which force holds neutrons and protons in the nucleus?

A) Magnetic B) Strong nuclear C) Gravity D) Coulombic

Answer: B. Strong nuclear force.

46. [L1] Rutherford's and Maxwell's statements are contradictory.

A) True B) False

Answer: A. Maxwell is correct.

47. [L1] Bohr said which is quantized?

A) Mass B) Angular momentum C) Temperature D) Charge

Answer: B. $mvr = nh/2\pi$.

48. [L1] Initial level of an emission is
    A) n_1 (lower) B) n_2 (higher) C) 1 D) ∞ always
    Answer: B. The transition starts at the higher level.
49. [L1] White light gives
    A) Line spectrum B) Continuous spectrum C) Absorption only D) None
    Answer: B. All wavelengths.
50. [Slides] Isobars are atoms with the same
    A) Protons B) Mass number C) Neutrons D) Electrons only
    Answer: B. Isotopes: same Z; isobars: same A; isotones: same N.
51. [Slides] The orbital is the region where the electron probability is about
    A) 10–20% B) 50% C) 90–95% D) 100%
    Answer: C. 3-D region of maximum probability.

Cheat sheet

Series: Lyman 1, Balmer 2, Paschen 3, Brackett 4, Pfund 5 (n_1 = lower, goes first).
Max E / min λ: from ∞. Min E / max λ: from n_1 + 1.
Bohr follow-ups: radius ↑, velocity ↓ with n; c = photon's speed; electron v < c; real path elliptical (Sommerfeld); mvr constant.
de Broglie: $\lambda = h/mv$; everything has λ (humans too, negligible).
Heisenberg: $\Delta x \cdot \Delta p \geq h/4\pi$; position certain → velocity uncertain.
Schrödinger: wave functions ψ; solving gives quantum numbers; ψ^2 = probability.

Nodes: radial n−l−1 · angular l · total n−1. 1s: none. s: no angular.
Quantum numbers: n energy level (1,2,3…) · l shape (0 to n−1; s p d f) · m_l orientation (−l…+l) · m_s spin (±½).
Count of m_l in a shell = n^2. Orbitals: s1, p3, d5, f7.
Impossible: 1p, 1d, 2d, 3f; n = 0; m_l outside −l…+l.
Config: Aufbau (low n+l first; tie → lower n) · Pauli (4 QNs unique; max 2 per orbital, opposite spin) · Hund (single first, same spin; half/full stable).
Exceptions: Cr $4s^1 3d^5$, Cu $4s^1 3d^{10}$.